

ACC_Ver01_CoreControl_Requirements

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Published on:14-Jul-2026

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Requirement Set: ACC_Ver01_CoreControl_Requirements

Description

1 System Initialization

Requirement Type Functional

ID ACC-REQ-001

Description

On system start, the controller shall enter the Power_Off with compressor_on = 0, fan_on = 0, fault_indicator = 0, and status outputs reset.

Links

Artifact [ACC_Ver01_CoreControl.slx](#)

 [Power Off](#) (↔Implemented by)

2 Temperature Setpoint Input

Requirement Type Functional

ID ACC-REQ-002

Description

The controller shall use set_temperature as the desired room temperature reference for cooling control.

Links

Artifact [ACC_Ver01_CoreControl.slx](#)

 [Power On](#) (↔Implemented by)

3 Power ON Command

Requirement Type Functional

ID ACC-REQ-003

Description

When power_button == 1, the controller shall enter the enabled operating state, compute temperature_error = room_temperature - set_temperature, and determine cooling demand based on temperature_error and temperature_hysteresis_band.

Links

Artifact [ACC_Ver01_CoreControl.slx](#)

 [Power On](#) (↔Implemented by)

 [Cooling Demand](#) (↔Implemented by)

 [\[power_button == 1 \]](#) (↔Implemented by)

4 Power OFF Command

Requirement Type Functional

ID ACC-REQ-004

Description

When power_button == 0, the controller shall enter the Power_Off state and set compressor_on = 0 and fan_on = 0.

Links

Artifact [ACC_Ver01_CoreControl.slx](#)

 [Power Off](#) (↔Implemented by)

 [\[power_button == 0 \]](#) (↔Implemented by)

5 Cooling Demand Activation

Requirement Type Functional

ID ACC-REQ-005

Description

If temperature_error >= temperature_hysteresis_band, the controller shall set cooling_demand = 1.

Links

Artifact [ACC_Ver01_CoreControl.slx](#)

 [Cooling Demand Activation](#) (←Implemented by)

 [\[temperature_error >= temperature_hysteresis_band \]](#) (←Implemented

by)

6 Cooling Demand Deactivation

Requirement Type Functional

ID ACC-REQ-006

Description

If temperature_error <= -temperature_hysteresis_band, the controller shall set cooling_demand = 0.

Links

Artifact [ACC_Ver01_CoreControl.slx](#)

 [Cooling Demand Deactivation](#) (←Implemented by)

 [\[temperature_error <= -temperature_hysteresis_band \]](#) (←Implemented

by)

7 Compressor Minimum OFF Time

Requirement Type Functional

ID ACC-REQ-007

Description

After the compressor turns OFF, the controller shall prevent compressor restart until compressor_min_off_time has elapsed.

Links

Artifact [ACC_Ver01_CoreControl.slx](#)

 [\[after\(compressor_min_off time, sec\) && cooling_demand == 1 \]](#)

(←Implemented by)

8 Compressor Activation

Requirement Type Functional

ID ACC-REQ-008

Description

If cooling_demand == 1 and the compressor is not in minimum-OFF-time lockout, the controller shall set compressor_on = 1.

Links

Artifact [ACC Ver01_CoreControl.slx](#)

 [\[after\(compressor min off time, sec\) && cooling_demand == 1 \]](#)

(Implemented by)

 [Compressor Activation](#) (Implemented by)

9 Compressor Deactivation

Requirement Type Functional

ID ACC-REQ-009

Description

If cooling_demand == 0, power_button == 0, or a fault condition is active, the controller shall set compressor_on = 0.

Links

Artifact [ACC Ver01_CoreControl.slx](#)

 [\[cooling_demand == 0 \]](#) (Implemented by)

 [\[power_button == 0 \]](#) (Implemented by)

 [\[fault == 1 \]](#) (Implemented by)

 [Compressor Deactivation](#) (Implemented by)

 [Power Off](#) (Implemented by)

 [Fault State](#) (Implemented by)

10 Fan Activation During Compressor Activation

Requirement Type Functional

ID ACC-REQ-010

Description

When compressor_on == 1, the controller shall set fan_on = 1.

Links

Artifact [ACC Ver01_CoreControl.slx](#)

 [Fan Activation](#) (Implemented by)

 [\[compressor_on == 1 \]](#) (Implemented by)

 [\[compressor_on == 1 \]](#) (Implemented by)

 [\[compressor_on == 1 \]](#) (Implemented by)

11 Fan Behavior After Compressor Deactivation

Requirement Type Functional

ID ACC-REQ-011

Description

When compressor_on == 0 after active cooling, the controller shall keep fan_on = 1 for configured fan_run_on_time, then set fan_on = 0.

Links

Artifact [ACC Ver01_CoreControl.slx](#)

 [\[compressor_on == 0 \]](#) (Implemented by)

 [Compressor Off Check](#) (Implemented by)

by)

 [\[after\(fan_run_on time, sec\) && compressor_on == 0 \]](#) ( Implemented

 [Fan Deactivation](#) ( Implemented by)

Appendix

Artifact List

Simulink models:

#	Name	Folder	Version
1	ACC_Ver01_CoreControl.slx	D:\MBD-SILMIL_May_2026\MBD-Requirements_Based_Stateflow_Projects\AC Controller Project\ACC-Ver01-CoreControl\model	1.1